Tony Zhang

guanyuzh@umich.edu TonyZhang2004 Tony Zhang (734) 730-1836

Education

University of Michigan – Ann Arbor **Ann Arbor, MI**

BSc. in Computer Science and Honors Mathematics (Pure Math), Major GPA 3.90/4 *Expected December 2026*

- **Selected Coursework: (*Graduate level)**
  Operating Systems, Distributed Systems, System Design of Search Engines, Machine Learning, Data Structures & Algorithms, Database Management Systems, Theory of CS, Computer Architecture, Probability Theory*, Discrete Stochastic Process*, Honors Analysis (Manifolds & Topology)
- **Extracurricular Activities**: Y Combinator Startup School, UMich Hackathon, Michigan ICPC Team, Michigan Data Science Team, UMich Quant Conference, Michigan Investment Group, Michigan Finance & Mathematics Society
- **Skills**: Python, C++/C, Rust, Git, React, AWS, MySQL, Linux, SQLite, Cuda, Node.js, Flask, FastAPI, Docker

Experience

Incoming Software Engineer **2027.02 (Expected)**

Old Mission Capital *Chicago, IL*

Software Engineering Intern **2026.06 – Present**

Ellipsis Labs *New York City, NY*

- Built Phoenix-01 Prediction Market Protocol smart contract infrastructure for fully collateralized binary options markets on Solana, enabling market creation, order placement, resolution, redemption, and protocol fee management in **Rust**
- Integrated Solana native Phoenix Central Limit Order Book liquidity with YES-token trading and synthetic NO position accounting, enforcing collateral invariants across limit orders, market orders, cancellations, and settlement flows
- Developed SDK/CLI tooling and a fully automated settlement crank that rolls crypto currency prediction markets, consumes Pyth price feeds, resolves expired markets, and concurrently force-cancels/redeems trader positions

Quantitative Developer **2024.09 – Present**

Michigan Investment Group *Ann Arbor, MI*

- Selected to compete in annual trading competition at Michigan and ranked **Top 10%** PnL among **over 200** contestants

Selected Awards

- **85th William Lowell Putnam Math Competition Top 400 individual nationwide (Top 7 in Michigan)**
- **Bronze Medal at Chinese Math Olympiad (Top 450 in Mainland China, as competitive as USAMO)**
- **Evelyn O. Bychinsky Award (selected 8 awardees among outstanding Math undergrads in Michigan)**

Publications/Manuscripts

ProoFit: Confidential Auditing of Model Overfitting **2026.01**

Joint with S. Sadhu, Prof. Amrita Roy Chowdhury, University of Michigan *Submitted to NeurIPS 2026*

Settling Weighted Token Swapping up to Algorithmic Barriers (arXiv preprint) **2025.07**

Joint with Prof. Nicole Wein, University of Michigan *Submitted to SOSA 2026*

Projects

Time Efficient Deep Learning Neural Networks Overfitting Detection Framework **2025.05**

- Developed a **PyTorch** framework to compare dropout-regularized vs. overfitting-prone neural networks on the German Credit dataset, leveraging **Pandas**, **Scikit-learn**, and **Matplotlib** for data preprocessing, scaling, and visualization
- Benchmarked model behavior across architectures including **ResNet18** and **VGG11** on **MNIST** using **Seaborn**
- Achieved a 28% reduction in overfitting, improving generalization gap from **18.2%** to **13.1%** by incorporating **dropout layers**, leveraging **LASSO penalty terms** and analyzing Frobenius norms of layer weights on deep learning networks

Citadel Securities Summer Invitational Datathon – Data Analysis on U.S. Food Industry **2024.08**

- Selected as **one of 120** participants from **over 1000** competent applicants across United States and Canada
- Performed **Exploratory Data Analysis** using **Pandas** and performed **feature engineering** to gain insightful observations on **correlation matrices** and **t-values** of **aggregated tabular** and **time-series** datasets
- Built a robust **LASSO regression** model using **Scikit-learn** to **resolve multi-collinearity** across features and facilitate **interpretability** of the model, resulted in an $R^2 = \mathbf{0.87}$ in predicting future ETFs and Commodities
- Designed an **XGBoost** model using our **synthetic features** which resulted in an **83%** pricing prediction accuracy by utilizing **regularization & early stopping** to prevent overfitting as well as improve generalization accuracy